

Tarea sección 2.6

3)

$$a) \lim_{x \rightarrow \infty} f(x) = -2$$

$$d) \lim_{x \rightarrow 3} f(x) = -\infty$$

$$b) \lim_{x \rightarrow -\infty} f(x) = 2$$

e) Ecuaciones de las asínt.: :

$$y = -2$$

$$x = 1$$

$$y = 2$$

$$x = 3$$

$$c) \lim_{x \rightarrow 1} f(x) = \infty$$

a) Trazar la gráfica:

$$f(0) = 3$$

$$\lim_{x \rightarrow 0^-} f(x) = 4$$

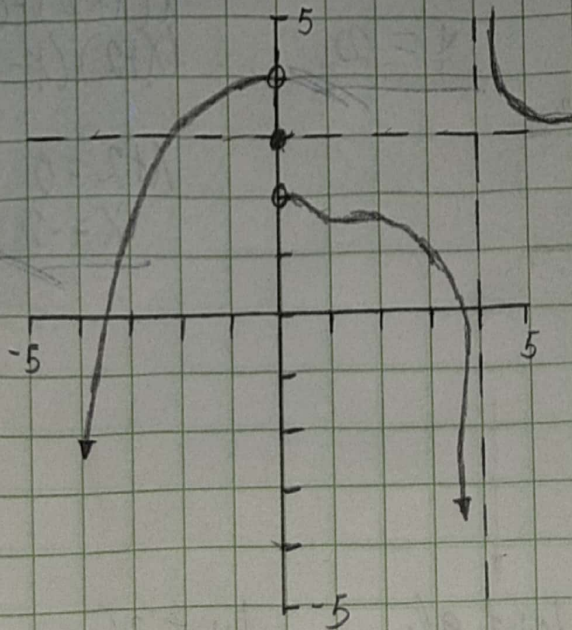
$$\lim_{x \rightarrow 0^+} f(x) = 2$$

$$\lim_{x \rightarrow -\infty} f(x) = -\infty$$

$$\lim_{x \rightarrow 4^-} f(x) = -\infty$$

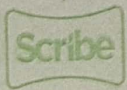
$$\lim_{x \rightarrow 4^+} f(x) = \infty$$

$$\lim_{x \rightarrow \infty} f(x) = 3$$



$$13) \lim_{x \rightarrow \infty} \frac{2x^2 - 7}{5x^2 + x - 3} = \frac{2x^2}{5x^2} = \frac{2}{5}$$

$$14) \lim_{t \rightarrow \infty} \frac{\sqrt{t} + t^2}{2t - t^2} = \frac{t^2(\sqrt{\frac{1}{t^3}} + 1)}{t^2(\frac{2}{t} - 1)} = \frac{\sqrt{\frac{1}{t^3}} + 1}{\frac{2}{t} - 1} = \frac{\sqrt{\frac{1}{\infty^3}} + 1}{0 - 1} = \frac{0 + 1}{-1} = -1$$



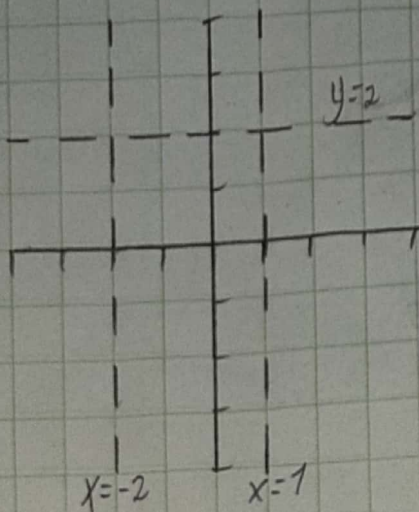
$$25) \lim_{x \rightarrow \infty} \frac{\sqrt{x+3x^2}}{4x-7} = \frac{x \sqrt{\frac{1}{x} + 3}}{x(4 - \frac{7}{x})} = \frac{\sqrt{\frac{1}{\infty} + 3}}{4 - \frac{7}{\infty}} = \frac{\sqrt{3}}{4}$$

$$35) \lim_{x \rightarrow \infty} \arctan(e^x) = \tan^{-1}(\lim_{x \rightarrow \infty} e^x) = \tan^{-1}(\infty) = \frac{\pi}{2}$$

$$47) \lim_{x \rightarrow \infty} [\ln(1+x^2) - \ln(1+x)] = \ln\left(\frac{1+x^2}{1+x}\right) = \ln\left[\lim_{x \rightarrow \infty} \left(\frac{1+x^2}{1+x}\right)\right]$$

$$\frac{1+\infty^2}{1+\infty} = \infty$$

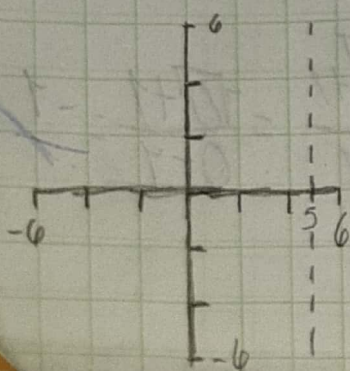
$$49) y = \frac{2x^2 + x - 7}{x^2 + x - 2}$$



Asint. Hb.
 $\frac{2}{1} = 2$
 $y = 2$

Asint. V.
 $x^2 + x - 2 = 0$
 $x(x+2) - (x-1) = 0$
 $(x+2)(x-1) = 0$
 $x+2=0 \quad x-1=0$
 $x=-2 \quad x=1$

$$51) y = \frac{x^3 - x}{x^2 - 6x + 5}$$



Asint. Hb.
 No existe

Asint. V.
 $x^2 - 6x + 5 = 0$
 $x(x-1) - 5(x-1) = 0$
 $(x-1)(x-5) = 0$

$x-1=0 \quad x-5=0$
 $x=1 \quad x=5$

Scribe